INCH-POUND

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SUPERSEDING
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# COMMERCIAL ITEM DESCRIPTION

METER, MODULATION
(150 kHz to 1.3 GHz)

The General Services Administration has authorized the use of this Commercial Item Description (CID) for all federal agencies

1. SCOPE. This Commercial Item Description (CID) describes a frequency modulation (FM), amplitude modulation (AM) and phase modulation (PM) modulation meter. This instrument gives a complete accurate characterization of modulated signals in the frequency range of 150 kHz to 1.3 GHz. This CID is meant as a minimum requirement for the modulation meter, in which only those manufacturers that meet or surpass the following requirements are supplied per this CID.

2. CLASSIFICATION. The equipment shall be capable of operation within the accuracies, limits, and specifications herein. Equipment covered by this CID may be commercially available equipment modified to the extent necessary to meet the following description. The equipment shall be Class 3 in accordance with MIL-PRF-28800 except as specified herein.

3. SALIENT CHARACTERISTICS.

3.1. General Description. The modulation meter shall be able to analyze characteristics of a modulated carrier with amplitude, frequency and phase modulation envelopes with a minimum frequency span from 150 kHz to 1.3 GHz. The modulation meter shall provide quantitative information of carrier level, noise figures and modulation, deviation and phase angle modulation levels in the carrier frequency.

Comments, suggestions, or questions on this document should be addressed to:
WR-ALC/404th SCMS/GUEEA, Robins AFB GA 31098-1813 or email SPEC99@us.af.mil
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AMSC N/A FSC 6625

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3.2. Frequency Range. The modulation meter shall analyze a carrier frequency with a minimum frequency span from 150 kHz to 1.3 GHz as stated in Table I.

**Table I. Frequency Bands**

| Band Name | ITU Band | Frequency | IEEE Designation |
|---|---|---|---|
| Low Frequency (LF) | 5 | 30~300 kHz | Ultra Sound |
| Medium Frequency (MF) | 6 | 300~3000 kHz | Hyper Sound |
| High Frequency (HF) | 7 | 3~30 MHz | HF* |
| Very High Frequency (VHF) | 8 | 30~300 MHz | VHF* |
| Ultra High Frequency (UHF) | 9 | 300~3000 MHz | UHF* (L &S Bands) |

3.3. Internal Time Base Reference Frequency. The modulation meter shall have a reference oscillator with a minimum frequency of 10 MHz.

3.4. Reference Frequency Aging . The modulation meter shall have a reference oscillator aging $\leq \pm 3 \bullet 10^{-6}$ /Yr. with an operating temperature compensation range from 0° to 50°C.

3.5. Carrier Frequency Resolution. The modulation meter shall have a minimum resolution of 10 Hz for carrier frequencies ≤ 1.0 GHz, and 100 Hz for carrier frequencies ≥ 1.0 GHz

3.6. Carrier Frequency Sensitivity. The modulation meter shall have a minimum input level sensitivity ranging from -40dBm to +27 dBm (2.25mV to 5 $V_{RMS}$) @ 50Ω, without pre-amplification.

3.7. Carrier Frequency Level Accuracy. The modulation meter shall have a input level accuracy variation of ≤ 4% throughout the minimum operating frequency band from 150 kHz to 1.3 GHz and a minimum input level sensitivity range from -40dBm to +27 dBm (2.25mV to 5 $V_{RMS}$).

3.8. Carrier Frequency Phase Noise. The modulation meter shall have a minimum input noise level measurement accuracy of ≤ ±2 dB to –125 dBc/Hz throughout the minimum operating frequency spectra 150 kHz to 1.3 GHz at input level of 0 dBm (0.2236 mV) @ 50Ω.

3.9. Input Impedance. The modulation meter shall have a typical frequency compensated input impedance of 50 Ω.

3.10. Input Connector. The modulation meter shall have an “N” type radio frequency connector for the input of signal.

3.11. Frequency Modulation Capture/Rejection. The modulation meter shall have a minimum bandwidth, deviation, variation, baseband and distortion (as seen in table II).



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3.12. Residual Noise. The modulation meter residual FM (noise) content shall have ≤ 20 Hz at 1.0 GHz attenuated at ≈10 dB per decade to ≤ 1 Hz at 100 MHz and below or ≤ -70dBm.

**Table II. Frequency Modulation Capture & Deviation**

| Function | Frequency Bands | | | |
|---|---|---|---|---|
| **Carrier Range** | 150kHz~3MHz | 3MHz~15MHz | 15MHz~400MHz | 400MHz~1.3GHz |
| **Deviation Range (± 10%)** | 0~9kHz | 0~120kHzkHz | 0~120kHz | 0~500kHz |
| **Deviation Accuracy Variation** | ≤ 2.5%@ 6kHz | ≤ 2%@ 6kHz | ≤ 2%@ 12kHz | ≤ 2%@ 250kHz |
| **Modulation Frequency (Typ)** | 30~12kHz | 30~48kHz | 30~48kHz | 30~48kHz |
| **AF Output Distortion** | ≤2%@35kHz | ≤2%@75kHz | ≤1%@100kHz | ≤1%@100kHz |

3.13. Signal to Noise and Distortion Ratio (SINAD). The modulation meter signal to noise and distortion ratio content shall be ≤ - 50 dBm.

3.14. Harmonic Distortion. The modulation meter shall have a demodulated frequency modulation distortion content of ≤ 2% of Total Harmonic Distortion (THD) at mid-band frequency of very high (VHF) and ultra-high (UHF) frequency or ITU bands 8 and 9.

3.15. Amplitude Modulation Envelope. The modulation meter shall have a amplitude modulation envelope or modulation measurement accuracy of ±2% from 0 to 100% modulation.

3.16. Amplitude Modulation Distortion. The modulation meter shall have a amplitude modulation envelope distortion measurement accuracy ≤ ±3% Total Harmonic Distortion (THD), combined SINAD (see table III).

**Table III. Noise Distortion and Modulation**

| Carrier Frequency (MHz) | Tolerance (%) + Digit Resolution by Band | | | | Modulation Level |
|---|---|---|---|---|---|
| | 30Hz~12kHz | 30~48kHz | 30~48kHz | 30Hz~48kHz | |
| 0.150~3 | ≤±2% + 1 | - | - | - | 0 ≤ M% ≤ 99 |
| 3~15 | - | ≤±2% + 1 | - | - | 0 ≤ M% ≤ 99 |
| 15~400 | - | - | ≤±2% + 1 | - | 0 ≤ M% ≤ 99 |
| 400~1,300 | - | - | - | ≤±2% + 1 | 0 ≤ M% ≤ 99 |



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3.17. Audio Filters. The modulation meter shall have user selectable high low/pass filters with de-emphasis time constants with typical values of 25, 50, 75, and 750 µS for accurate measurement of modulated content, noise filtering and de-masking per Table IV.

**Table IV. Filter Frequency Roll Off**

| Typical Filter Frequency | Typical Cut Off Frequency dB Roll $dB_{(Typical)}$ |
|---|---|
| 20Hz~60Hz | -3,12,18&24 |
| 250Hz~300Hz | -3,12,18&24 |
| 2,500kHz~3,000Hz | -3,12,18&24 |
| 15kHz~20kHz | -3,12,18&24 |
| 48kHz~50kHz | -3,12,18&24 |

3.18. Phase Modulation. The instrument shall have phase modulation accuracy of ± 3% of reading ±1 digit for modulation rates from 200 Hz to 20 kHz, with phase shifts up to 400 radians per Table V.

**Table V. Phase Modulation Data**

| Function | Frequency Bands | | | |
|---|---|---|---|---|
| **Carrier Range** | 150kHz~3MHz | 3MHz~15MHz | 15MHz~400MHz | 400MHz~1.3GHz |
| **Phase Shift Rads (± 10%)** | 0~9Rad | 0~120Rad | 0~120Rad | 0~500Rad |
| **Deviation Accuracy Variation** | ≤3.5%@ 6 Rad/s | ≤ 3.5%@ 6 Rad/s | ≤ 3.5%@ 12 Rad/s | ≤ 3.5%@ 250 Rad/s |
| **Modulation Frequency (Typ)** | 60~12kHz | 60~48kHz | 30~48kHz | 30~48kHz |
| **AF Output Distortion** | ≤2%@30 Rad/s | ≤2%@75 Rad/s | ≤2%@100 Rad/s | ≤2%@100 Rad/s |

3.19. Audio Output. The modulation meter shall have an audio signal output with output impedance of 600Ω or 10kΩ capable of providing 0dB ($0.7746V_{RMS}$).

3.20. Intermediate Frequency Output. The modulation meter shall have an IF output. The output voltage shall be ≥ -27dBm ($10mV_{RMS}$) into a 50 ohms load.

3.21. Demodulated Signal Output. The modulation meter shall have a demodulated baseband signal output capable of providing 0dB ($0.7746V_{RMS}$).

3.22. Communications. The modulation meter shall be compliant with the general purpose interface bus (GPIB) protocol, IEC-60488-1 & IEC-60488-2 for connectivity, programmability, command, control and data transfer.

3.23. System Safety. The modulation meter shall comply with all sections of the Underwriters Laboratories (UL), IEC/ISO standard 61010-1 for electrical safety.

3.24. Weight. The system shall comply with MIL-STD-1472 section 5.9.11 and table XLIV of section 5.9.17.3.4 for weight & size restrictions for single person lift & handling.



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3.25. Sound Levels. The Modulation Meter acoustic noise shall conform to sound level requirements on section 1 of ISO 7779 measured per section 2 of ISO 8253-2.

3.26. Electrostatic Discharge (ESD). The design of the modulation meter shall preclude equipment damage due to ESD, protect personnel from electrical shock due to static charging, and prevent ignition of explosive atmospheres due to sparking in accordance with MIL-STD-1686 section 5.2.2.3.

3.27. Electro Magnetic Interference (EMI). The modulation meter electro-magnetic interference (EMI) emissions shall be in accordance with the following radiated emission and susceptibility requirements of MIL-PRF-28800, MIL-STD-461: RE102 and RS103 & IEC-1000-4 thru IEC-1000-6. Emitted radiation absorption and radiation limits FCC Part 15.109, Part 15.209 and IEC-61000 sub volume 4-3.

3.28. Service Life. The design of the modulation meter shall be such that under normal use and operation the service repair can be performed according to MIL-PRF-28800 sections 3.14.2 & 3.14.3.

3.29. Port Damage. The input and output port(s) shall have a protection method that can be achieved by an active protection circuit or fuses either internal or external.

3.30. Connectors. The system shall have a "N" type connector for the input signal and BNC type connectors, for other input & output functions. A General Purpose Interface Bus (GPIB) interface IEC-60488 Protocol connector is required for compatibility and control functions. The connectors shall comply with MIL-PRF-39012 section 3 for requirements.

3.31. Operating Temperature Range. The modulation meter shall meet the specifications listed herein with full capabilities within the temperature range of 32°F to 122°F (0°C to 50°C) after tests in MIL-PRF-28800 section 4.5.5.1.1 have been performed for class 3 equipment classification (Per table 11 on page 41).

3.32. Storage Temperature Range. The modulation meter shall meet and/or exceed manufacturers specifications for stored equipment after tests of MIL-PRF-28800 section 4.5.5.1.1 temperature, humidity & storage have been performed for instruments in the range of class 3 equipment classification (Per table 11 on page 41).

3.33. Altitude Requirements. The modulation meter shall meet its operating performance and accuracy while operating at an altitude of 8,000 ft. (2,438 meters). The equipment shall meet its performance and accuracy requirements, for operation and non-operating storage, as specified in MIL-PRF-28800 section 3.8.3 and section 4.5.5.2 respectively at this altitude.

3.34. Heat Dissipation and Ventilation. Ventilation shall be sufficient to allow for prevention, buildup and retention of moisture, heat dissipation either passive or active shall be suitable in order to meet MIL-PRF-28800 section 4.5.5.1.1 for class 3 equipment classification (Per table 11 on page 41).



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3.35. Calibration and Maintenance Adjustments. The design of the modulation meter shall provide readily accessible calibration and maintenance adjustments. The calibration adjustments shall be accessible without removal of the test set case or modules. No special tools, software, technical manuals or documentation shall be required for the alignment, adjustment, repair and calibration of the modulation meter except those provided with the test set. The calibration interval shall be based on a period of one calendar year or an operating time of 2,080 hours, whichever occurs first. At the end of this interval 85% of the units shall remain within calibration tolerances in accordance with MIL-PRF-28800 section 3.12.

3.36. Safety and Environmental. The equipment shall meet all safety and environmental requirements as specified in MIL-PRF-28800 for the classification as stated herein.

3.37. Reliability. The design of the modulation meter shall be such that under normal use and operation the mean time before failure (MTBF) exceeds 2,080 hours of operation with a statistical certainty equal or greater than 95% in accordance with MIL-PRF-28800 section 3.13. The manufacturer shall provide a summary of maintenance or warranty records to establish the validity of the statistical assertion in accordance with MIL-PRF-28800 section 3.13. The manufacturer shall be able to reasonably establish the claim that the offered equipment is capable of the required level of reliability by meeting and/or exceeding MIL-PRF-28800 sections 3.13 and section 4.5.8.1.

3.38. Electrical Power Sources and Connections. The equipment shall operate from nominal commercial, military, and shipboard power source of 120Vrms or 240Vrms single-phase AC (± 10 %) at line frequencies of 50 Hz and 60 Hz (± 5 %) as specified in MIL-PRF-28800.

## 4. REGULATORY REQUIREMENTS.

4.1 Recycled recovered materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with 23.403 of the Federal Acquisition Regulation (FAR). However, used, rebuilt, or refurbished items shall not be provided.

4.2 Green Procurement Program. Green Procurement Program (GPP) is a mandatory federal acquisition program that focuses on the purchase and use of environmentally preferable products and services. GPP requirements apply to all acquisitions using appropriated funds, including services and new requirements. FAR 23.404(b) applies and states the GPP requires 100% of EPA designated product purchase that are included in the Comprehensive Procurement Guidelines list that contains recovered materials, unless the item cannot be acquired: a) competitively within a reasonable timeframe; b) meet



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appropriate performance standards, or c) at a reasonable price. The prime contractor is responsible for ensuring that all subcontractors comply with this requirement.

## 5. PRODUCT CONFORMANCE PROVISIONS

5.1 Product Conformance. The products provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The government reserves the right to require proof of such conformance before first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

5.2 Metric Products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the specified tolerances using conversion tables contained in the latest revision of ASTM-D2260, and all other requirements of this CID are met.

## 6. PACKAGING.

Preservation, packing, and marking shall be as specified in the contract or order.

## 7. NOTES.

7.1 Sources of Documents.

7.1.1 Military Specifications, Standards, and Handbooks referenced herein may be obtained from the Standardization Documents Order Desk, 700 Robbins Ave, Bldg 4, Section D, Philadelphia, PA 19111-5094 or online at http://quicksearch.dla.mil .

7.1.2 The Code of Federal Regulation (CFR) may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC, 20402 or online at http://www.gpo.gov/ .

7.1.3 The ISO/IEC standards copies of these documents are available online at http://www.iec.ch/ or IEC Regional Centre for North America (IEC-ReCNA) 446 Main Street, 16th Floor Worcester, MA 01608 USA.

7.1.4 The FAR may be obtained from the Superintendent of Documents, P.O. Box 371954, Pittsburgh PA 15250-7954. Electronic copies of the FAR may be obtained from https://www.acquisition.gov/far/.

7.1.5 ASTM documents may be obtained online at http://www.astm.org/ or from ASTM, 100 Barr Harbor Drive, P O Box C700, West Conshohocken, PA 19428.



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7.2 National Stock Number NSN. The following is a list of NSNs assigned that correspond to this CID. The list may not be indicative of all possible NSNs associated with this CID.

| NSN | NSN |
|---|---|
| 6625-01-079-3394 | 6625-01-243-7904 |
| 6625-01-097-6010 | 6625-01-513-3405 |
| 6625-01-154-4844 | |

7.3 Key Words

Amplitude Modulation (AM)
Frequency Modulation (FM)
Harmonic
Phase Modulation (PM)

7.4 Changes from previous issue. Marginal notations are not used in this in this revision to identify changes with respect to the previous issue due to the extent of the changes.

MILITARY INTEREST

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Air Force - 84

Preparing Activity:
Air Force - 84

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